

CARLOS MORENO ROJAS

Biomedical Engineering · Electrical Engineering · Research & Development

Interdisciplinary engineer with experience in biomedical instrumentation, electronics, embedded systems, organ perfusion and preservation, medical imaging, biomedical optics, robotics, FPGA, CAD and experimental validation. Current Master's research focuses on ex vivo liver perfusion and ICG-NIR functional assessment.

EDUCATION

MSc Biomedical Engineering

Pontificia Universidad Católica de Chile · 2025–2027

Research in ex vivo liver perfusion, biomedical instrumentation and ICG-NIR.

Electrical Engineering

Pontificia Universidad Católica de Chile · 2025–2026

Electronics, embedded systems, data acquisition, control and FPGA.

Biomedical Engineering

Pontificia Universidad Católica de Chile · 2021–2024

Bachelor's Degree in Engineering Sciences - Biomedical Engineering.

RESEARCH & DEVELOPMENT

Master's Research - Liver Viability & ICG-NIR

Development of an instrumented ex vivo liver perfusion platform; physiological baseline characterization; pressure, flow, temperature and vascular resistance monitoring; ICG-NIR fluorescence kinetics and functional parameter extraction.

Borealis UC - Organ Supercooling

Research on organ preservation, supercooling and thermal control.

Organ Preservation Machine

Electromechanical, fluidic and control-system redesign and miniaturization.

SELECTED ENGINEERING PROJECTS

CYBATHLON 2024 - Ciervo UC

Electrical Area Leader

Electrical and electromechanical integration of a robotic transfemoral prosthesis for CYBATHLON 2024 in Zürich.

Lung Segmentation - U-Net

Medical image segmentation using deep learning.

Universal Testing Machine - FPGA

Zybo Z10, Vivado and VHDL for acquisition and processing of mechanical-test variables.

Vscan Air Enclosure

CAD and prototyping for a portable ultrasound accessory.

TECHNICAL SKILLS

- Biomedical instrumentation, physiological sensing and experimental validation
- Python, C++, signal processing and image processing
- ESP32, STM32, Arduino, PCB and embedded systems
- Zybo Z10, Vivado and VHDL
- NIR fluorescence, biomedical optics and ICG
- Fusion 360, CAD, 3D printing and mechanical prototyping

TEACHING & LEADERSHIP

Physics Laboratory Teaching Assistant

Pontificia Universidad Católica de Chile

FIS154.

Biomedical Engineering Student Chapter UC

President · 2024–2025

IEEE EMBS PUC Chile

Webmaster · since 2024

LANGUAGES

- Spanish - Native
- English - B2 / ALTE 3